BabyVision Experiments: RL Fine-Tuning of Gemma-4 for Visual Reasoning

Yoav Sinai

Contents

1 Abstract

This report documents a seminar project built on top of the BabyVision benchmark, a visual reasoning benchmark of 388 puzzle-style tasks spanning fine-grained discrimination, spatial perception, visual tracking, and pattern recognition. The original paper demonstrates that reinforcement learning (RL) fine-tuning can improve a model's visual reasoning performance on this benchmark. This project applies that same RL recipe to `google/gemma-4-E4B-it`, a model not covered in the original paper, and evaluates whether the approach transfers.

The best configuration (multimodal LoRA targets with an increased correctness reward weight) improved accuracy from a 12.97% baseline to 13.75%. During the course of the project, an evaluation-script defect was identified and corrected: Gemma's generation was being capped at a substantially smaller token budget than other models under evaluation, causing a large share of its answers to be truncated before completion. A separate issue also came to light: an earlier reported Qwen baseline result had been carried forward from an

undocumented, broken evaluation run and did not reproduce - once re-run under the current pipeline, Gemma and Qwen turned out to perform comparably as untrained baselines, overturning the conclusion that Gemma clearly outperformed Qwen presented in class earlier in the project. Across the project, the token generation budget consistently affected whether a model produced an extractable answer at all - cutting the no-answer rate by up to 26 points - while its effect on final accuracy was real but considerably more modest, moving accuracy by roughly 1-4 points depending on configuration, comparable in size to the effect of the RL training design choices evaluated. A secondary evaluation of ByteDance's BAGEL-7B-MoT on the generation track initially suggested a nonzero result on the benchmark's hardest subtype, but manual verification revealed this was a scoring error by the automated LLM judge rather than a genuine correct answer, underscoring that such judges cannot be relied on without spot-checking, particularly on visually dense discrimination tasks.

2 Introduction

The objective of this project was to evaluate whether RL fine-tuning, as proposed in the BabyVision paper, generalizes to a model not evaluated in the original work. `google/gemma-4-E4B-it` was selected as the target model. `Qwen/Qwen2.5-7B-Instruct` was used throughout as an automated judge, comparing each model's extracted answer against the ground truth. A stock, untrained `Qwen/Qwen2.5-VL-7B-Instruct` was evaluated in parallel as a baseline comparison point.

All results reported below are based on a small-sample evaluation protocol (388 tasks, 3 passes per configuration) appropriate for a seminar-scale investigation. The results should be interpreted as directional evidence of what helps, not as a statistically rigorous benchmark claim.

2.1 Project Structure

This section is a navigation aid to the repository, oriented toward finding this report's supporting evidence rather than describing every file. Paths are relative to the repository root.

```
BabyVision/
  results/
    final_report.pdf                          <- this report
    accuracy_comparison.png                   <- chart used in Results

  babyvision_eval/                            <- MLLM (text-answering) track
    evaluate_model.py, compute_score.py,      <- original fork (API-based only)
    utils.py
    evaluate_local_hf.py                      <- local-GPU evaluation (Gemma + Qwen),
                                                  added this project
    evaluate_system_prompt.py                 <- two-stage system-prompt evaluation
    rl_train_gemma4.py                        <- RL training, Configuration B
    rl_train_gemma4_correctness_weight2.py    <- RL training, Configuration C (best)
    rl_train_system_prompt.py                 <- RL training, Configuration D
    runs/                                     <- evaluation outputs, one subfolder per
                                                  configuration x token budget, e.g.
                                                  runs/rl_correctness_weight2_maxtok1024/
                                              (model_results_run_*.json, score_summary.txt)

  babyvision_gen_eval/                        <- generation (image-editing) track
    scripts/
```

```
    inference_bagel.py                    <- BAGEL native image-editing inference
    evaluate_local_hf.py                  <- local-GPU judge for this track
 generated/bagel_find_different/round1/  <- BAGEL outputs, Find-the-different test
  results/
   bagel_find_different/round1/eval.jsonl <- judged per-task results for that test
    summary.txt, summary.json              <- full 280-task run, all models
    summary_find_different.txt/.json      <- dedicated summary for the 16-task test

sbatch_scripts/
  training/, evaluation/, other/          <- one SLURM script per job referenced
                                             in Appendix A

logs/
  training/, evaluation/, setup/, archive/ <- SLURM stdout/err, named by job ID
                                               (see Appendix A for the mapping)

data/
  babyvision_data/meta_data.jsonl         <- the 388-task MLLM benchmark
  babyvision_gen_data/                    <- the generation-track benchmark
```

Every job ID referenced in this report (Appendix A) can be traced to its exact stdout/err pair under `logs/`, and every accuracy number in Section 5 traces back to the raw per-task JSON under the corresponding `babyvision_eval/runs/*/` directory.

2.2 Codebase Provenance

The forked repository (`babyvision_eval/`) originally provided only an API-based evaluation script (`evaluate_model.py`), a scoring utility (`compute_score.py`), and shared helper functions (`utils.py`). No local-GPU evaluation path and no RL training code existed in the original fork. The following components were implemented for this project (paths relative to `babyvision_eval/`, except the final two rows, which are under `babyvision_gen_eval/scripts/`):

Script	Purpose
`evaluate_local_hf.py`	Local-GPU evaluation for Gemma and Qwen (the upstream script supports only API-based inference). This is where the token-budget defect described below originated.
`evaluate_system_prompt.py`	Two-stage evaluation for the system-prompt RL variant: one model call generates a system prompt, and a second, frozen call answers the question using that prompt.
`rl_train_gemma4.py`	GRPO training with correctness_weight=1.0 and language-model-only LoRA targets (Configuration B).
`rl_train_gemma4_correctness_weight2.py`	GRPO training with correctness_weight=2.0 and multimodal LoRA targets (Configuration C, the best-performing setup).

Script	Purpose
`rl_train_system_prompt.py`	GRPO training for the system-prompt-generating policy (Configuration D).
`inference_bagel.py`	Runs ByteDance's BAGEL-7B-MoT in its native image-editing mode on the generation track. An earlier iteration of this project evaluated BAGEL through the text-answering understanding track instead, which was determined to be an inappropriate benchmark for this model; see the BAGEL discussion below and the deprecated understanding-track notes for details.
`evaluate_local_hf.py` (generation track)	Local-GPU judge for the generation track (the upstream `evaluate.py` supports only API-based judging). This script is model-agnostic by design and required no modification to judge BAGEL's output.

3 Method

Four training/evaluation configurations were evaluated in sequence, in addition to an untrained baseline:

- **Configuration A (Baseline).** Stock Gemma, no training.
- **Configuration B (RL, correctness_weight=1.0).** GRPO training with LoRA adapters applied only to language-model layers.
- **Configuration C (RL, correctness_weight=2.0).** GRPO training with two modifications relative to Configuration B: LoRA adapters were extended to the multimodal connector and projection layers, and the correctness component of the reward was doubled, reducing the relative incentive to satisfy the output-format reward without producing a correct answer.
- **Configuration D (System-prompt RL).** Rather than training the model to answer directly, this configuration trains the model to generate a system prompt for itself, on the hypothesis that a self-generated instruction could improve downstream answer quality.

4 Identification and Resolution of an Evaluation Defect

During initial analysis, the seminar instructor noted that Qwen's performance on the benchmark looked anomalous. Separately, a defect was identified in `evaluate_local_hf.py`: Gemma's generation was capped at 256 tokens, while every other model under evaluation, including Qwen, received a 512-token budget. This asymmetry caused a substantial fraction of Gemma's answers to be truncated before the model could produce a final answer within the required `\boxed{...}` format. The defect was corrected by applying a uniform budget across all models, and the budget was subsequently raised further, to 1024 tokens, to minimize truncation entirely. Every configuration was re-evaluated at each budget; all headline results in this report use the 1024-token evaluation unless noted otherwise.

Separately, an earlier reported Qwen baseline result at 512 tokens (2.84% accuracy, 81.4% no-answer rate), unverifiable against any retained job record, did not reproduce: re-running it under the current pipeline (job `23820791`) produced a very different result - 12.54% accuracy, 3.9% no-answer rate - under an identical, confirmed 512-token budget. Matched individual outputs showed the earlier run's generations were truncated

after roughly 150-170 tokens, well short of 512, indicating it came from a broken, undocumented earlier version of the evaluation script, not from a genuine limitation of the model. The earlier figure has been superseded by the reproduction throughout this report, and once a correctly-run Qwen baseline is used, the conclusion that Gemma clearly outperformed Qwen presented in class no longer holds (see Results below).

Increasing the token budget substantially reduced the no-answer rate for every configuration (by up to 26 points between 256 and 1024 tokens), but its effect on accuracy was considerably more modest (roughly 1-4 points over the same range). This indicates that the budget primarily determines whether an answer is captured at all, not what the model is capable of getting right: it recovers attempts that would otherwise be truncated, but most of those recovered attempts are still wrong.

5 Results

5.1 Final Results (1024-token budget)

Configuration	Accuracy	No-answer rate
A: Baseline Gemma	12.97% ± 0.99%	6.6% ± 0.2%
B: RL (correctness_weight=1, language-only LoRA)	11.34% ± 0.56%	6.7% ± 0.6%
C: RL (correctness_weight=2, multimodal LoRA)	**13.75% ± 0.49%**	0.3% ± 0.2%
D: System-prompt RL	13.66% ± 0.56%	8.3% ± 1.5%
Qwen baseline (untrained)	13.32% ± 0.44%	1.9% ± 0.3%

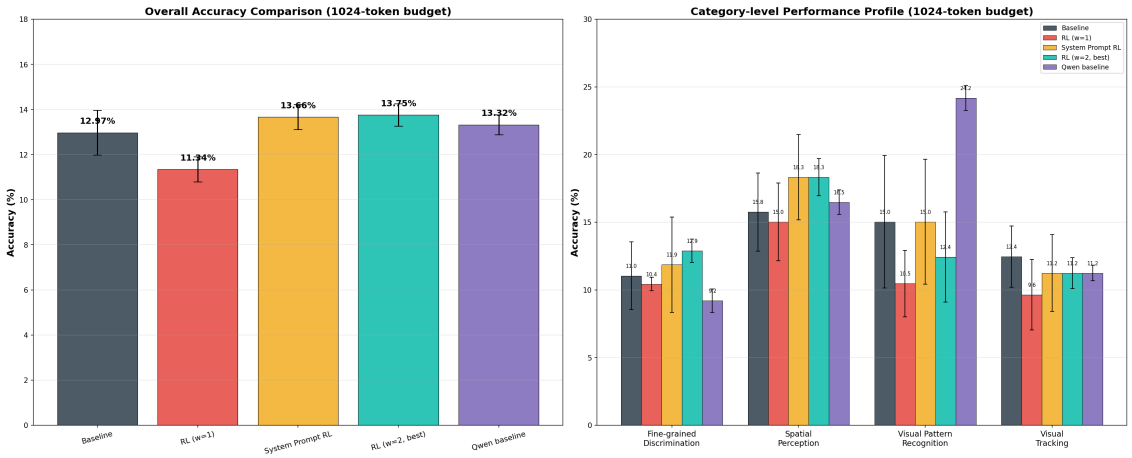
Several observations follow from these results:

- Configuration C and the Qwen baseline are approximately tied for the best result, with Configuration D close behind. The range between the best and worst configuration (11.34% to 13.75%) is considerably narrower than it appeared at earlier, truncation-affected token budgets, indicating that a portion of the apparent differences between configurations reflected differential sensitivity to truncation rather than differences in visual reasoning capability.
- Configuration B performs marginally worse at 1024 tokens than it did at 512 tokens (12.20% versus 11.34%). No definitive explanation for this regression was established; it may reflect sampling variance given the limited evaluation set (388 tasks, 3 passes), or a behavioral characteristic of this specific checkpoint under a larger generation budget. This result is reported without further interpretation, given the available evidence.
- Configuration C maintains the lowest no-answer rate of any configuration at every token budget tested, a consistent effect attributable to the increased correctness-reward weighting, which appears to train the model toward producing complete, concise answers.
- With a uniform 1024-token budget and no training applied to either model, Gemma (12.97%) and Qwen (13.32%) achieve comparable accuracy. Neither model demonstrates a clear advantage on this benchmark under equivalent evaluation conditions.
- Configuration D's result (13.66%) is close to Configuration C's despite a different architectural rationale (a self-generated system prompt rather than a directly reward-shaped answer policy). One speculative explanation, offered here only as a hypothesis and not as a conclusion this project supports, is

that Configuration D's improvement over baseline reflects the additional inference-time compute from its two model calls per task rather than a benefit specific to the system-prompt-generation architecture itself. This project does not include a compute-matched control that would isolate the two explanations, so no claim is made either way.

5.2 Category-Level Results

Category	Baseline	RL (B)	RL (C)	System-prompt RL (D)	Qwen
Fine-grained Discrimination	11.04%	10.43%	12.88%	11.86%	9.20%
Spatial Perception	15.75%	15.02%	18.32%	18.32%	16.48%
Visual Pattern Recognition	15.03%	10.46%	12.42%	15.03%	**24.18%**
Visual Tracking	12.45%	9.64%	11.24%	11.24%	11.24%



Qwen's Visual Pattern Recognition result (24.18%) is the strongest single-category result across all configurations evaluated and warrants further investigation beyond the scope of this project.

5.3 Subtype-Level Results (Baseline vs. Best RL Configuration)

All 22 subtypes are listed below, in the same order they appear in each run's `score_summary.txt` (categories in the log's fixed order - Fine-grained Discrimination, Visual Tracking, Spatial Perception, Visual Pattern Recognition - and subtypes alphabetical within each category), so the table can be checked directly against the log files referenced at the end of this section.

Category	Subtype	Baseline	RL (C)
Fine-grained Discrimination	2D Pattern Completion	33.33% ± 6.24%	**45.00% ± 4.08%**
Fine-grained Discrimination	Count Clusters	11.11% ± 4.54%	**12.96% ± 2.62%**
Fine-grained Discrimination	Count Same Patterns	1.90% ± 1.35%	1.90% ± 1.35% (tie)
Fine-grained Discrimination	Find the different	0.00% ± 0.00%	0.00% ± 0.00% (tie)
Fine-grained Discrimination	Find the same	1.96% ± 2.77%	1.96% ± 2.77% (tie)
Fine-grained Discrimination	Find the shadow	5.80% ± 4.10%	**7.25% ± 2.05%**
Fine-grained Discrimination	Pattern and Color Completion	30.00% ± 8.16%	**33.33% ± 2.36%**
Fine-grained Discrimination	Reconstruction	**7.14% ± 5.83%**	2.38% ± 3.37%
Visual Tracking	Connect the lines	7.02% ± 2.48%	**15.79% ± 0.00%**
Visual Tracking	Lines Observation	0.00% ± 0.00%	0.00% ± 0.00% (tie)
Visual Tracking	Maze	**20.00% ± 4.08%**	18.33% ± 2.36%
Visual Tracking	Metro map	**5.56% ± 3.93%**	0.00% ± 0.00%
Visual Tracking	Recognize numbers and letters	**18.84% ± 4.10%**	11.59% ± 2.05%
Spatial Perception	3D Cube Unfold	**5.56% ± 3.93%**	0.00% ± 0.00%
Spatial Perception	3D Pattern Completion	33.33% ± 9.07%	**46.30% ± 6.93%**
Spatial Perception	3D Views	11.11% ± 3.02%	11.11% ± 3.02% (tie)
Spatial Perception	Count 3D blocks	13.64% ± 3.71%	**15.15% ± 2.14%**
Spatial Perception	Paper Folding	13.89% ± 10.39%	**16.67% ± 11.79%**
Visual Pattern Recognition	Logic Patterns	**4.76% ± 6.73%**	2.38% ± 3.37%
Visual Pattern Recognition	Mirroring Patterns	**10.00% ± 0.00%**	6.67% ± 4.71%
Visual Pattern Recognition	Overlay Patterns	**21.57% ± 7.34%**	15.69% ± 2.77%
Visual Pattern Recognition	Rotation Patterns	23.33% ± 12.47%	**26.67% ± 20.55%**

The largest gains for Configuration C over baseline occurred in 3D Pattern Completion (33.33% → 46.30%) and Connect the Lines (7.02% → 15.79%). The most notable regressions occurred in Recognize numbers and letters (18.84% → 11.59%) and Overlay Patterns (21.57% → 15.69%). The largest subtype-level regressions for Configuration C overall were 3D Cube Unfold and Metro map, which each declined from 5.56% at baseline to 0.00%. Of the 22 subtypes, 5 were exact ties between the two configurations (Count Same Patterns, Find the different, Find the same, Lines Observation, 3D Views).

Raw per-task outputs for all configurations, all passes, are available in `model_results_run_*.json` under each configuration's output directory; the subtype-level numbers above can be reproduced di-

rectly from `score_summary.txt` in `babyvision_eval/runs/baseline_maxtok1024/` and `babyvision_eval/runs/rl_correctness_weight2_maxtok1024/`.

6 Discussion

6.1 Mechanism Underlying the Configuration C Improvement

The improvement from Configuration B to Configuration C is attributable to two concurrent changes, both of which appear necessary:

- **Multimodal LoRA target modules.** Configuration B applied LoRA adapters only to language-model layers. The available evidence is consistent with this producing a representational drift between the model's text output and the frozen vision encoder's representations, since only the language-side parameters were updated. Extending LoRA to the multimodal connector and projection layers in Configuration C appears to preserve this alignment.
- **Correctness reward weighting.** Under correctness_weight=1.0, the formatting component of the reward (correct use of `\boxed{...}`) was inexpensive to satisfy relative to producing a correct answer, permitting the policy to accumulate reward through format compliance with limited regard for correctness. Doubling the correctness weight in Configuration C shifted a larger proportion of the gradient signal toward answer correctness.

These findings suggest that GRPO training improves visual reasoning performance on this benchmark under two conditions: LoRA adaptation must extend to the multimodal connector layers, and the correctness reward must be weighted sufficiently to prevent the policy from optimizing primarily for output format.

6.2 The Generation Token Budget as a Confounding Variable

Table 1 below summarizes accuracy and no-answer rate across all three token budgets evaluated during this project.

Table 1: Accuracy and no-answer rate by token budget.

Configuration	Acc. (256 tok)	Acc. (512 tok)	Acc. (1024 tok)	No-answer (256)	No-answer (512)	No-answer (1024)
A: Baseline Gemma	11.08%	12.03%	12.97%	24.4% ± 0.5%	16.8% ± 2.1%	6.6% ± 0.2%
B: RL (weight=1)	9.54%	12.20%	11.34%	27.2% ± 1.5%	16.7% ± 1.4%	6.7% ± 0.6%
C: RL (weight=2)	13.14%	13.66%	13.75%	2.1% ± 0.4%	0.7% ± 0.3%	0.3% ± 0.2%
D: System-prompt RL	9.71%	10.48%	13.66%	34.5% ± 1.8%	21.1% ± 2.3%	8.3% ± 1.5%
Qwen baseline	-	12.54% ± 0.44%	**13.32%**	-	3.9% ± 0.6%	1.9% ± 0.3%

The Qwen baseline has no 256-token entry because the original token-cap defect (see "Identification and Resolution of an Evaluation Defect") applied only to Gemma; Qwen was evaluated at 512 tokens or higher throughout.

Two distinct effects are visible in Table 1, and they are not the same size. The effect on the no-answer rate is large and consistent: moving from 256 to 1024 tokens reduces it by 17.8 points for Configuration A, 20.5 points for Configuration B, and 26.2 points for Configuration D (Configuration C, whose reward already selects for concise answers, starts low and only drops a further 1.8 points). The reproduced Qwen baseline shows the same direction of effect at a smaller scale: a 2.0-point drop in no-answer rate between 512 and 1024 tokens.

The effect on accuracy is real but considerably smaller. Over the same 256-to-1024 range, accuracy moves by 1.89 points for Configuration A, 1.80 points for Configuration B, 0.61 points for Configuration C, and 3.95 points for Configuration D; the reproduced Qwen baseline moves by 0.78 points between 512 and 1024 tokens. In other words, giving the model room to finish an answer overwhelmingly converts a non-extractable, automatically-incorrect response into a genuine attempt, but most of those newly-completed attempts are still wrong: the token budget determines whether an answer gets captured, not what the model is fundamentally capable of getting right. Judged on accuracy, its effect was of comparable size to, rather than clearly larger than, the effect of the RL training design choices evaluated - so all comparisons in this report use a fixed, generous (1024-token) budget across every configuration.

Training used a separate, fixed completion budget rather than the evaluation-time token budget discussed above: the RL training scripts (`rl_train_gemma4.py`, `rl_train_gemma4_correctness_weight2.py`) generated rollouts with `max_completion_length=256`, and `rl_train_system_prompt.py` used `max_completion_length=128`, uniformly across all training runs regardless of configuration. Whether training rollouts were themselves frequently truncated at this budget, and whether that would have affected the quality of the learned policy, was not investigated and is noted here as an open question for follow-up work.

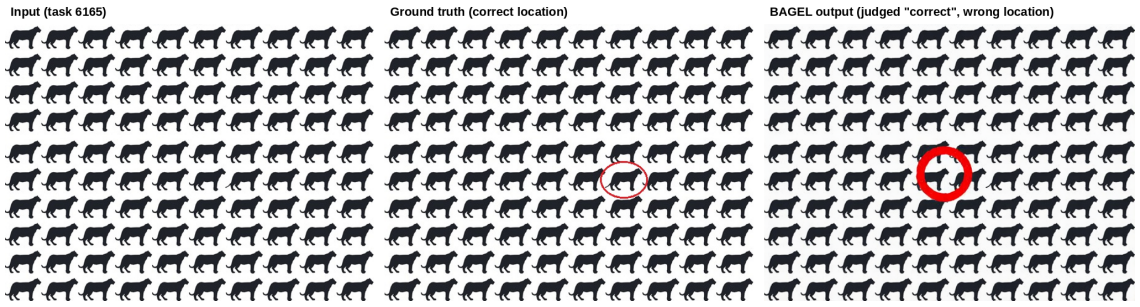
A related observation is that Configuration C did not actually need the larger evaluation-time token budget that this section is otherwise about. Its no-answer rate was already low even at the smallest budget tested - 2.1% at 256 tokens, versus 24.4% for the untrained baseline at the same budget - and only improved marginally as the budget increased (0.7% at 512, 0.3% at 1024; see Table 1). In other words, Configuration C was already answering within a much smaller budget than the one ultimately used; the 1024-token cap benefited the other configurations far more than it benefited Configuration C. This is consistent with GRPO training under the higher correctness weight pushing the policy toward more concise, direct answers rather than longer chains of reasoning, fitting a reward structure that under correctness_weight=2.0 more heavily favors producing a correct `\boxed{...}` answer than favors extended deliberation on the way there.

6.3 Implications for BAGEL Evaluation

ByteDance's BAGEL-7B-MoT was evaluated on the generation track because the original BabyVision paper cites it by name as an example of unified understanding-and-generation models that "offer a new path forward" for solving tasks through image-space reasoning rather than text. A full 280-task, multi-round evaluation of BAGEL was judged impractical given the compute cost of running its generation pipeline at scale, so the evaluation was instead narrowed to the single hardest available case: the subtype "Find the different," the only subtype on which every other evaluated model in this project (Configurations A through D and the Qwen baseline) scored exactly 0%. BAGEL was evaluated on the 16 available tasks of this subtype using its native image-editing capability (marking the target region directly on the input image, rather than producing a text answer). The automated judge (Qwen2.5-VL, comparing the input, ground truth, and generated images)

scored 1 of the 16 responses correct.

That single “correct” result does not hold up under manual inspection. Comparing the exact pixel location of the circle in the judged-correct response (task 6165) against the ground truth shows they mark different animals in the grid - the judge’s vision-based comparison was fooled by the task’s dense, highly repetitive layout, where verifying the precise location of a circled element is a harder visual discrimination problem than it may appear.



With that single case corrected, BAGEL’s true score on this subtype is 0 of 16 (0%), the same as every other configuration evaluated in this project. No configuration - text-based or image-based - solved any task in the hardest subtype tested. The example above is retained not as a positive result but as a reminder that automated LLM-based judges are not fully reliable on visually dense discrimination tasks, and that a “correct” score from such a judge merits spot-checking before being reported, particularly when it is the only positive data point supporting a claim.

This generation-track evaluation followed an earlier, incorrect attempt: BAGEL was initially evaluated through the text-answering understanding track, producing an overall score of 4.12% with a 78% rate of responses truncated before completion of the model’s internal reasoning process. That evaluation is considered to have targeted an inappropriate benchmark for a model of this architecture, since BAGEL’s relevant capability is native image editing rather than text-based answering, and is retained for reference in `babyvision_eval/_deprecated/bagel_understanding_track/` but is not treated as part of the primary results of this project.

7 Conclusion

RL fine-tuning of `google/gemma-4-E4B-it` using the BabyVision paper’s recipe produced a modest but consistent improvement over baseline (13.75% versus 12.97%), attributable to the combination of multimodal LoRA targeting and increased correctness reward weighting. Two methodologically significant issues were identified and resolved along the way: an evaluation-script defect that under-budgeted Gemma’s generation relative to other models, and a broken, undocumented earlier evaluation run that had produced an unreliable Qwen baseline figure. Once both were corrected, the conclusion that Gemma clearly outperformed Qwen presented in class earlier in the project no longer held - Gemma and Qwen perform comparably as untrained baselines, and the RL-tuned Gemma configuration and the untrained Qwen baseline are approximately tied for the best overall result obtained in this project.

The generation token budget mattered most for exactly the configurations whose no-answer rate it moved the most. Configurations A, B, and D and the Qwen baseline all had double-digit no-answer rates at smaller budgets, and a larger budget consistently made more of their answers extractable at all; Configuration C, whose no-answer rate was already low even at 256 tokens, barely changed with a larger budget because there were few truncated answers left to recover. The relationship between a lower no-answer rate and higher

accuracy was not perfectly monotonic for every configuration - Configuration B's accuracy at 1024 tokens is slightly below its 512-token value despite a further drop in no-answer rate (see Results) - but the overall pattern holds: the token budget is not a property of the model being evaluated, but of the evaluation itself, and it must be held fixed and generous across every configuration being compared, or differences in no-answer rate alone can masquerade as differences in capability.

A secondary evaluation of BAGEL-7B-MoT on the generation track's hardest subtype ("Find the different") initially reported a nonzero result, but manual inspection showed this was a judge error - the automated LLM judge (Qwen2.5-VL) was fooled by the task's dense, repetitive layout into scoring a wrongly-located annotation as correct. BAGEL's true score is 0 of 16, matching every text-based configuration on this subtype. The practical lesson carries beyond this one result: automated LLM-based judging, used throughout this project to score both the text-answering and generation tracks, is not infallible on visually demanding discrimination tasks, and a single positive result - especially one that is the sole basis for a claim - warrants manual verification before being reported.

Appendix A: Job IDs and Logs (for reproducibility)

Run	Job ID	Logs
Baseline eval (256 tok, original)	`16875224`	Out / Err
RL training, w=1	`16875257`	Out / Err
RL eval, w=1 (256 tok, original)	`16876673`	Out / Err
RL training, w=2	`16877068`	Out / Err
RL training, w=2, follow-up (no-op - training had already finished)	`16877168`	Out / Err
RL eval, w=2 (256 tok, original)	`16877072`	Out / Err
System-prompt RL training	`16877070`	Out / Err
System-prompt RL eval (256 tok, original)	`16877197`	Out / Err
Baseline eval, 512 tokens	`23604080`	Out / Err
RL eval, w=1, 512 tokens	`23657919`	Out / Err
RL eval, w=2, 512 tokens	`23657920`	Out / Err
System-prompt RL eval, 512 tokens	`23657921`	Out / Err
Qwen baseline eval, 512 tokens (reproduction, verified)	`23820791`	Out / Err
Baseline eval, 1024 tokens	`23671051`	Out / Err
RL eval, w=1, 1024 tokens	`23671052`	Out / Err
RL eval, w=2, 1024 tokens	`23671053`	Out / Err
System-prompt RL eval, 1024 tokens	`23671054`	Out / Err
Qwen baseline eval, 1024 tokens	`23671055`	Out / Err

Run	Job ID	Logs
BAGEL “Find the different” eval, gen track	`23820802`	Out / Err